

The growth-promoting effect of water extract of *Chuanminshen violaceum* stem and leaf on broilers

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ABSTRACT Currently, developing nonantibiotic growth promoters is a broad consensus in broiler industry, which is one of the effective ways to reduce drug-resistant strains. *Chuanminshen violaceum* is a traditional Chinese medicinal herb that is commonly used for its roots, while the stems and leaves are often discarded, resulting in a huge amount of waste. This study optimized the preparation process of water extract of *Chuanminshen violaceum* stems and leaves (CVSLE) by response surface analysis based on the yields of polysaccharide and protein. The CVSLE and herbal powder (CVSL) were then processed into granules before being used as feed additives. The *Macleaya cordata* powder was used as positive control. The results showed that the addition of CVSLE (0.5% of the feed) showed the highest growth-promoting activity than other CVSLE

groups (0.2% and 1%), 1% CVSL group and positive control (0.05%). CVSLE at the dosage of 0.5% could significantly increase the ADG and reduce the FCR from d 21 to 42, d 0 to 42. The HI antibody titers against Newcastle disease virus and avian influenza virus were significantly enhanced at 21, 28 and 42 d. CVSLE did not affect the slaughtering performances, but could significantly elevate the spleen, thymus and bursa of Fabricius indices and the transcriptional levels of IL-2, IL-4, IL-10 and IFN- γ in spleen. The intestinal barrier function of broilers was significantly enhanced by increased levels of immune barrier (sIgA), physical barrier (ZO-1, OCL and Muc-2) and flora barrier (*Lactobacillus* and *Bifidobacterium*). These results suggest that CVSLE was a promising herbal additive candidate for broilers.

Key words: *Chuanminshen violaceum* stem and leaf, Herbal feed additive, Nonantibiotic growth promoter, broiler

2024 Poultry Science 103:104235

<https://doi.org/10.1016/j.psj.2024.104235>

INTRODUCTION

In modern livestock raising industry, feed additives, especially in-feed antibiotics, are widely used to reduce feeding costs and to meet the requirements of rapid turn-out (Biswas et al., 2010). In addition to treating bacterial infections, antibiotics have been developed as antibiotic growth promoters (AGP) which were widely used to promote livestock growth. However, AGP have caused serious negative effects, such as development of drug-resistant bacteria, drug residues, toxic side effects and environmental pollution (Salim et al., 2018). The European Union (EU) has completely stopped using all

AGP in EU member states since 2006, and the Ministry of Agriculture and Rural Affairs of China issued an announcement on July 10, 2019 to implement the policy to stop the production and sale of all AGP except for Traditional Chinese Medicine (TCM) since January 1, 2020 (Millet and Maertens, 2011). Currently, only 2 herbal extracts were approved in China, *Macleaya cordata* extract and the mixed extract of *Lonicera confusa* DC. and *Scutellaria baicalensis* Georgi. It was a tendency to shift the development goal of the livestock raising industry from single high-yield model to green, healthy, safe and pollution-free model. Thus, TCM exhibits great application prospects in phytogenic feed additives, which have the advantages of being multifunctional, low toxicity and side effects, and fewer residues (Kalia et al., 2018).

Chuanminshen violaceum is a plant of the genus *Chuanminshen* of the family Apiaceae, mainly distributed in Sichuan and Hubei province of China. Its root is a widely used medicinal and edible plant called “Chuan Min Shen” in Chinese, and traditionally used to

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Received June 18, 2024.

Accepted August 14, 2024.

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moisten the lungs, resolve phlegm, and nourish the stomach (He et al., 2022). Researches have shown that the main chemical constituents of the root are polysaccharides and coumarins, and also contain a small amount of flavonoids, steroids and triterpene acids, which have pharmacological effects such as antioxidant (Fan et al., 2017; Lin et al., 2019; Zou et al., 2023), expectorant (Song et al., 2013), cough suppressant (Zhao et al., 2015), immune enhancement and antimutagenic (Feng et al., 2015). The overground parts (stems and leaves) of *Chuanminshen violaceum* (CVSL) are the nonmedicinal parts, which have not been utilized yet. The annual production of *Chuanminshen violaceum* in China is about 100,000 tons, resulting in a great waste. The nonmedicinal part usually has the same active substances, which has the potential of medicinal usage. Accordingly, the objective of this study was to assess the potential growth-promoting properties of an aqueous extract derived from the stems and leaves of *Chuanminshen violaceum*, with the aim of developing a novel herbal growth promoter.

MATERIALS AND METHODS

The Preparation of Aqueous Extract of CVSL

CVSL was provided by Damingyuan *Chuanminshen violaceum* Growers' Co-operative (Bazhong, China) and identified taxonomically by Dr. Lixia Li (Sichuan Agricultural University, Chengdu, China). The herbal was dried, crushed, and then decocted with water. The decoction was centrifuged at 4,000 r/min for 15 min, and the supernatant was collected and concentrated by vacuum rotary evaporation at 80°C to obtain the aqueous extract of CVSL (CVSLE) (Dong et al., 2016; Lin et al., 2020). To optimize the extraction process, a combination of 1-way experiments and response surface analysis was used. The contents of total polysaccharides and proteins of the extract were used as the reference standards, which were measured by Phenol-sulfuric acid and Coomassie brilliant blue G-250 methods, respectively (Figure 1A).

Firstly, a 1-way experiment was used to determine the optimal range of influencing factors, including powder/

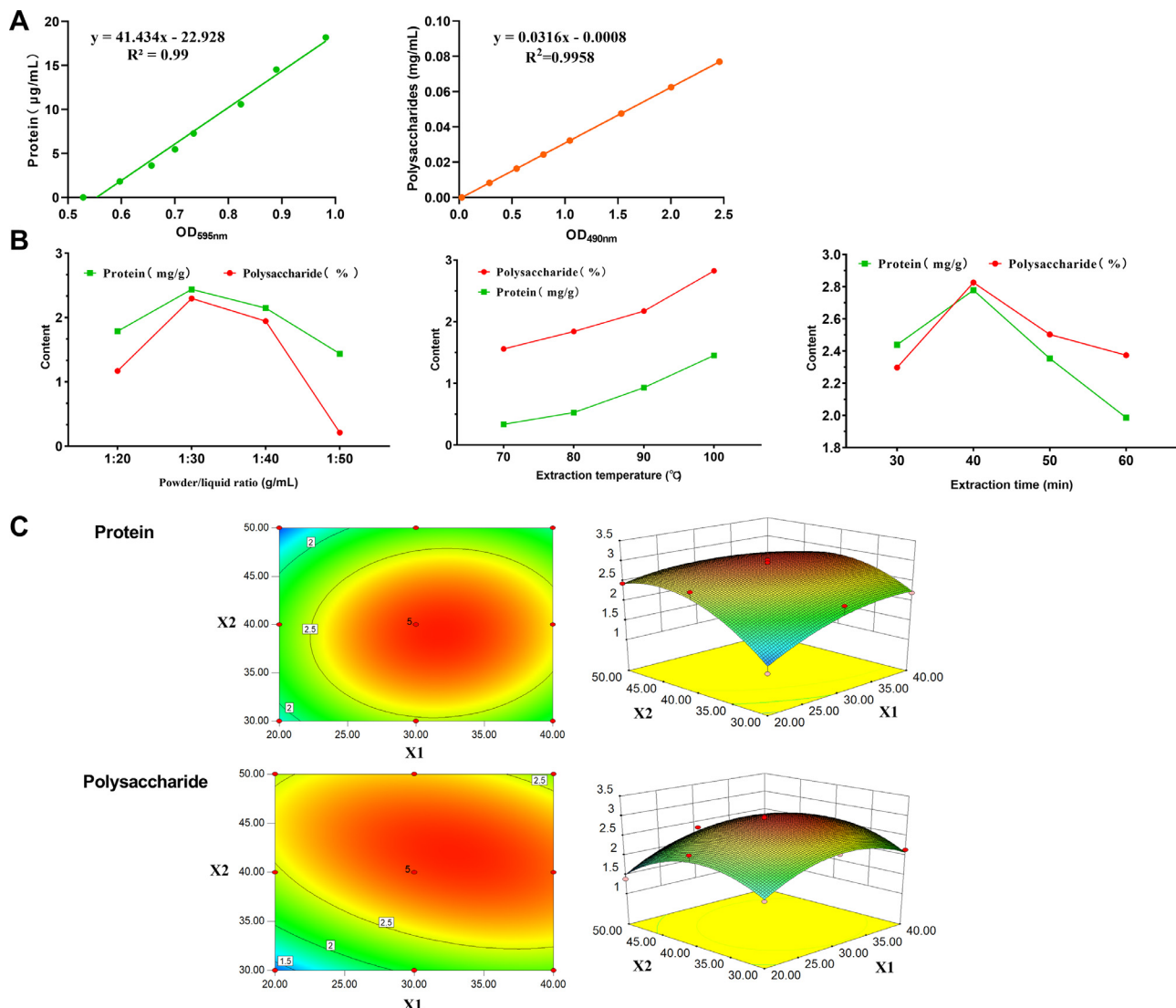


Figure 1. The extraction process of CVSLE. (A) The standard curve of protein and polysaccharide. (B) The 1-way experiments of powder/liquid ratio, extraction temperature and extraction time. (C) The 3-D response surface and the contour plots of protein and polysaccharide.

liquid ratio (g/mL; 1:20, 1:30, 1:40 and 1:50), extraction temperature (70, 80, 90, and 100°C) and time (30, 40, 50, and 60 min). The results (Figure 1B) showed that the polysaccharide and protein yield of the extract was significantly affected by the powder/liquid ratio (X1), extraction time (X2). The extraction temperature was set as 100°C. Then, a 2-factor, 3-level Plackett-Burman factorial response surface optimization was designed using the Design Expert software (Trial Version 8.0.6, Stat-Ease Inc.) to determine the best combination of these extraction variables. A total of 13 experimental points for optimizing the 2 individual parameters was shown in Table 1. After analysis of the experimental data by multiple regression fitting using Design Expert software, and the resulting mathematical equations were obtained: Protein yield = $2.93 + 0.18X_1 - 0.11X_2 + 0.057X_1X_2 - 0.49X_1^2 - 0.58X_2^2$; Polysaccharides yield = $2.92 + 0.20X_1 + 0.29X_2 - 0.25X_1X_2 - 0.27X_1^2 - 0.56X_2^2$. Then, based on the mathematical models, the 3-D response surface and the contour plots were generated to determine the optimal conditions (Figure 1C), which were as follows: extraction time 40.47 min and powder/liquid ratio 32.39 with the predicted yields of 2.95807% polysaccharides and 2.93432 mg/g proteins. To validate the suitability of the predicted optimum response values, 3 independent replications were conducted using the optimal conditions as extraction time 40 min, powder/liquid ratio 32 and extraction temperature 100 °C. The yields of polysaccharides and proteins were 2.89699% and 2.88307mg/g, respectively, which was not significantly different ($P > 0.05$) from the predicted value, suggesting that the optimal extraction process obtained from response surface is valid.

The Preparation of CVSL and CVSLE Granules

The CVSL and CVSLE was crushed and passed through an 80-mesh sieve. The powder was mixed with excipients of starch and dextrin in the ratio of 1:1:1, and water was used as a wetting agent to form granules. After being dried at 60°C, the granules passed through a 10-mesh sieve but did not pass through the an 80-mesh sieve. According to the 2020 edition of the Chinese

Pharmacopoeia, the particle size, moisture and weight loss of the CVSL and CVSLE granules meet the requirements of the pharmacopoeia.

Experimental Design, Diets, and Birds Management

The experiment was conducted using a single-factor design. The 180 male 1-day-old yellow-feathered broilers of similar weights were purchased from Zhengda Livestock and Poultry Co., Ltd. (Chengdu, China). All chickens were randomly divided into 6 treatment groups with 3 replicates per group, each consisting of 10 chickens. The chickens were fed and raised in cages for 6 wk (Mussa et al., 2022). The treatments included: blank control (Con), 0.05% *Macleaya cordata* powder (Positive control, PC; Hunan Mecoda Bio-Resources Co., China), 1% CVSL granules (CVSL), 0.2% CVSLE granules (CVSL-L), 0.5% CVSLE granules (CVSLE-M) and 1.0% CVSLE granules (CVSLE-H). The granules were added to the basal diet. The process of feeding broilers was divided into 2 phases: the early phases (1–21 d) was fed to newborn broiler feeds and the later phases (22–42 d) was fed to medium broiler feeds (Zerehdaran et al., 2009). The nutritional and composition of 2 feeds were in accordance with Formula feeds for layers and broilers (GB/T 5916-2020 of China) (Skinner-Noble and Teeter, 2003). The composition and nutritional levels of the experimental feeds are shown in Table 2.

The breeding and management of the animals in this experiment were in accordance with Regulations on the Management of Laboratory Animals (China) and all procedures involving animals and their care in this study were approved (No. 2022-0157) by the Ethics Committee of Sichuan Agricultural University. The temperature was controlled at 33-35°C for the first 1-3 d, not lower than 27°C for the first 3 wk, and then gradually decreased by about 3°C per week in the 4th and 5th wk, and maintained at around 20°C after the fifth wk. Lighting was provided for 24 h during the first 7 d and then gradually transitioned to 12 h of lighting. Broilers were given food and water at regular intervals during the

Table 1. Plackett-Burman design.

Number	Factors	Coded levels		
		Level -1	Level 0	Level +1
1	Powder/liquid ratio (g/ mL)	20	30	40
2	Extraction time(min)	30	40	50
Test number	X1	X2	Protein yield(mg/g)	Polysaccharides yield(%)
1	20	30	1.78729	1.1784
2	30	30	2.43907	2.29866
3	40	30	2.14781	2.20859
4	20	40	2.41699	2.62498
5	30	40	2.77833	2.85446
6	40	40	2.55373	2.75606
7	20	50	1.3818	2.44428
8	30	50	2.35436	2.50285
9	40	50	1.96866	2.47179
10	30	40	2.84619	2.78038
11	30	40	2.94223	2.87974
12	30	40	2.98743	3.02651
13	30	40	2.96385	2.96116

Table 2. Ingredient composition of basal diets.

Ingredients	Percentage(%)	
	1–21 d	22–42 d
Corn	60.94	66.49
Soybean meal	33.90	26.98
Soybean oil	1.62	2.71
Stone powder	1.47	1.52
CaHPO ₄	0.96	0.76
NaCl	0.32	0.40
Choline chloride	0.08	0.08
Antifungal agent	0.04	0.04
Lysine sulfate	0.07	0.25
DL-Methionine	0.15	0.26
Threonine	0.00	0.06
Premix	0.45	0.45
Total	100.00	100.00

Premix (per kg of diet): From 1 to 21 d in the experimental period: Fe 100 mg, Zn 100 mg, Mn 120 mg, Cu 8 mg, Co 1.0mg, Se 0.30 mg, Vit A 8000 IU, Vit D 1000 IU, Vit E 20 IU, Vit K 0.50 mg, Vit B₁ 2.0 mg, Vit B₂ 8 mg, Vit B₆ 3.5 mg, Vit B₁₂ 0.01 mg, Folic acid 0.55 mg, Biotin 0.18 mg, Niacin 35 mg, Pantothenic acid 10 mg, Choline 1300 mg, Crude protein 229g.

From 21 to 42 d in the experimental period: Fe 80 mg, Zn 80 mg, Mn 100 mg, Cu 8 mg, Co 1.0 mg, Se 0.30 mg, Vit A 6000 IU, Vit D 750 IU, Vit E 10 IU, Vit K 0.50 mg, Vit B₁ 2.0 mg, Vit B₂ 5 mg, Vit B₆ 3.0 mg, Vit B₁₂ 0.01 mg, Folic acid 0.55 mg, Biotin 0.15 mg, Niacin 30 mg, Pantothenic acid 10 mg, Choline 1000 mg, Crude protein 205g.

experiment. On d 7, inactivated vaccines (Strain La Sota + Strain M41 + Strain SS/94) against Newcastle disease, Infectious bronchitis and Avian influenza (Sub-type H9) were injected subcutaneously into the neck of the broiler. The remaining feeding management followed the Feed safety evaluation-Code of practice for the feeding trial of broilers (GB/T 40942-2021 of China).

Growth Performance

On the 1st, 21st, 28th and 42nd d of rearing, feed intake and body weight were recorded for each replicate after 12 h of broiler fasting and average daily gain (ADG), average daily feed intake (ADFI) and feed consumption ratio (FCR) were calculated for each period (Skinner-Noble and Teeter, 2004).

ADG = Weight gain/day

ADFI = Feed consumption/day

FCR = Feed consumption/Weight gain

Sample Collection

On the 21st, 28th and 42nd d of rearing, 2 chickens were randomly selected from each replicate of each group (n = 6), followed by blood sample collection (5 mL) from the jugular vein. Then, the broilers were exsanguinated by a severing of the jugular vein and carotid artery on 1 side of the neck. After dissection, the organs, including spleen, thymus, bursa, jejunum and ileum were collected for following assays. The serum was obtained by centrifugation at 3,000 r/min for 15 min after coagulation for 30 min at room temperature.

Slaughtering Performance

After bleeding and plucking, the weight of each broiler was recorded. The broilers were then eviscerated, and the semi-eviscerated carcass, eviscerated carcass, thigh, breast and abdominal fat were weighed, respectively (Rizzi et al., 2007; Biswas et al., 2021).

Dressed percentage = Slaughter weight/Live weight × 100%

Percentage of half – eviscerated yield

= Half – eviscerated weight/Live weight × 100%

Percentage of eviscerated yield = Eviscerated weight
/Live weight × 100%

Percentage of breast muscle

= Breast weight/Eviscerated weight × 100%

Percentage of leg muscle = Large and lower leg muscle weight
/Eviscerated weight × 100%

Percentage of abdominal fat = Abdominal fat

/(Eviscerated weight + Abdominal fat) × 100%

Hemagglutination Inhibition Assay

Serum samples from broilers in each group were taken for haemagglutination inhibition (HI) assays to evaluating antibody titers against Avian influenza virus (AIV) and Newcastle disease virus (NDV) (Pedersen, 2014). A 2-fold serial dilution of serum was made in a 96-well, V-shaped bottom microtiter plate. The dilution ranged from 1 : 2 to 1 : 2,048. Next, 50 µL of NDV and AIV antigen [4 Hemagglutination (HA) units] was added to all wells except for the last row, which served as the controls. The antigen-serum mixture was then incubated at 37°C for 10 min. Then, 50 µL of 1% chicken erythrocyte suspension was added to each well, followed by a 30-min incubation period. Agglutination was monitored and recorded, and the highest dilution of serum causing complete inhibition was considered the endpoint. The titer was expressed as reciprocal log2 values of the highest dilution that displayed HI.

Determination of the Immune Organ Index

The immune organs, including spleen, thymus and bursa were weighed after slaughter to calculate their immune organ index.

Immune organ index = Immune organ weight
/body weight × 100%

Real-Time PCR

Total RNA of spleen, jejunum and ileum tissue was extracted with the TRzol agent according to the manufacturer's instructions (No.RA101-01, Biomed, Beijing, China). The total RNA was reverse transcribed into cDNA using the M-MLV 4 First-Strand cDNA Synthesis Kit (No.MT403-01, Biomed, Beijing, China). The real-time PCR was performed to evaluate the gene expression levels using the Hieff UNICON Universal Blue qPCR SYBR Green Master Mix according to the manufacturer's instructions (No.11184ES03, Yeasen, Shanghai, China). The primer sequences used in this study are listed in Table 3. The results were calculated using the $2^{-\Delta\Delta C_t}$ method (Jozefczuk and Adjaye, 2011).

Microbiological Analysis

The cecum contents were diluted with phosphate buffer and incubated on agar plates for counting. *Lactobacillus* (LAB) were selected for cultured using LAB selection medium, *Bifidobacterium* using Bismuth Sulfite agar medium, *Escherichia coli* using Eosin Methylene Blue Agar medium and *Salmonella* using Salmonella Shigella (SS) agar medium (Clavijo and Flórez, 2018). *E. coli* and *Salmonella* were cultured aerobically at 37°C for 24 h, while *Lactobacillus* and *Bifidobacterium* were cultured anaerobically at 37°C for 48 h. Bacterial counts are expressed in lg CFU/g (Cesare et al., 2019).

Statistical Analysis

The data analysis was conducted using SPSS 20.0. One-way ANOVA were performed to test for significant differences between treatments. The results were expressed as mean pooled standard errors of the means (mean $\pm$ SEM). The probability value (P) < 0.05 was considered statistically significant.

Table 3. Primers used for Real-time PCR.

Gene	Primer(5'-3')	Tm/°C
β -actin	F:GTGACCTGACGGACTACCTC R:TCTCCTGCTCGAAATCCAGT	58.3
IL-2	F:GCTAATGACTACAGCTTATGGAGCA R:TGGGTCTCAGTTGGTGTGTAGAG	58.3
IL-4	F:GCTCTCAGTGCCGCTGATG R:GGAAACCTCTCCCTGGATGTC	58.3
IL-10	F:AGCAGATCAAGGAGACGTTT R:ATCAGCAGGTACTCCTCGAT	58.3
IFN- γ	F:ATCATACTGAGCCAGATTGTTTCG R:TCTTTACCTTCTTCACGCCAT	58.3
OCL	F:ATCGCCTCCATCGTCTACATC R:GCTGCACATGGCCAACAAG	56.3
ZO-1	F:TATGAAGATCGTGCGCTCC R:GAGGTCTGCCATCGTAGCTC	53.9
SIgA	F:GTCACCGTCACCTGGACTACA R:ACCGATGGTCTCCTTCACATC	56.3
Muc-2	F:TCACCCTGCATCGTACTTGCTCA R:TGTCCATCTGCCTGAATCAGGT	53.9

RESULTS

Growth Performance

The impact of dietary CVSL and CVSLE supplementation on growth performance is illustrated in Figure 2. In comparison to the blank control group, the ADG of the high-dose CVSLE group was found to be significantly higher from d 0 to 21 (P < 0.05). Furthermore, the ADG of all groups was observed to be significantly higher from d 0 to 42 and d 21 to 42 (P < 0.05). The ADG of the CVSLE group was higher than that of the CVSL group (P < 0.05), and it was comparable to that of the positive control group. No significant differences were observed in ADFI between the CVSLE and blank control groups. Conversely, the ADFI in the positive control group was significantly reduced in comparison with the blank control group from d 0 to 42 and d 21 to 42 (P < 0.05). The ADFI in the low-dose CVSLE group was significantly higher than that in the CVSL group from d 0 to 42 (P < 0.05). From d 0 to d 21, there were no statistically significant differences in FCR across all groups (P > 0.05). Significant differences were observed among all the groups from d 0 to 42 and d 21 to 42 (P < 0.05), with the FCR in the positive control group exhibiting the lowest level. The medium-dose and high-dose CVSLE groups demonstrated a similar FCR, which was significantly lower than that observed in the CVSL group (P < 0.05).

Slaughtering Performance

The impact of dietary CVSL and CVSLE supplementation on slaughtering performance is illustrated in Figure 3. With respect to slaughter rate, the high-dose CVSLE group exhibited a statistically significant increase in comparison to the positive control group (P < 0.05). In terms of total net chamber rate, the low and medium doses of the CVSLE groups demonstrated a statistically significant increase in comparison to the positive control group. Furthermore, the low-dose CVSLE group exhibited a statistically significant increase in comparison to the CVSL group (P < 0.05). In the hamstrings rate, the positive control group exhibited a statistically significant increase compared to both the blank control group and the medium-dose CVSLE group (P < 0.05). In abdominal fat rate, the low-dose CVSLE group and the positive control group exhibited significantly elevated rates in comparison to the blank group and the high-dose CVSLE group (P < 0.05). Additionally, the positive control group demonstrated significantly higher rates than the CVSL group and the medium-dose CVSLE group (P < 0.05). No notable discrepancies were observed among all groups in half-eviscerated yield and breast muscle rates (P > 0.05).

HI Antibody Titers

The impact of dietary CVSL and CVSLE supplementation on antibody titers against NDV and AIV is

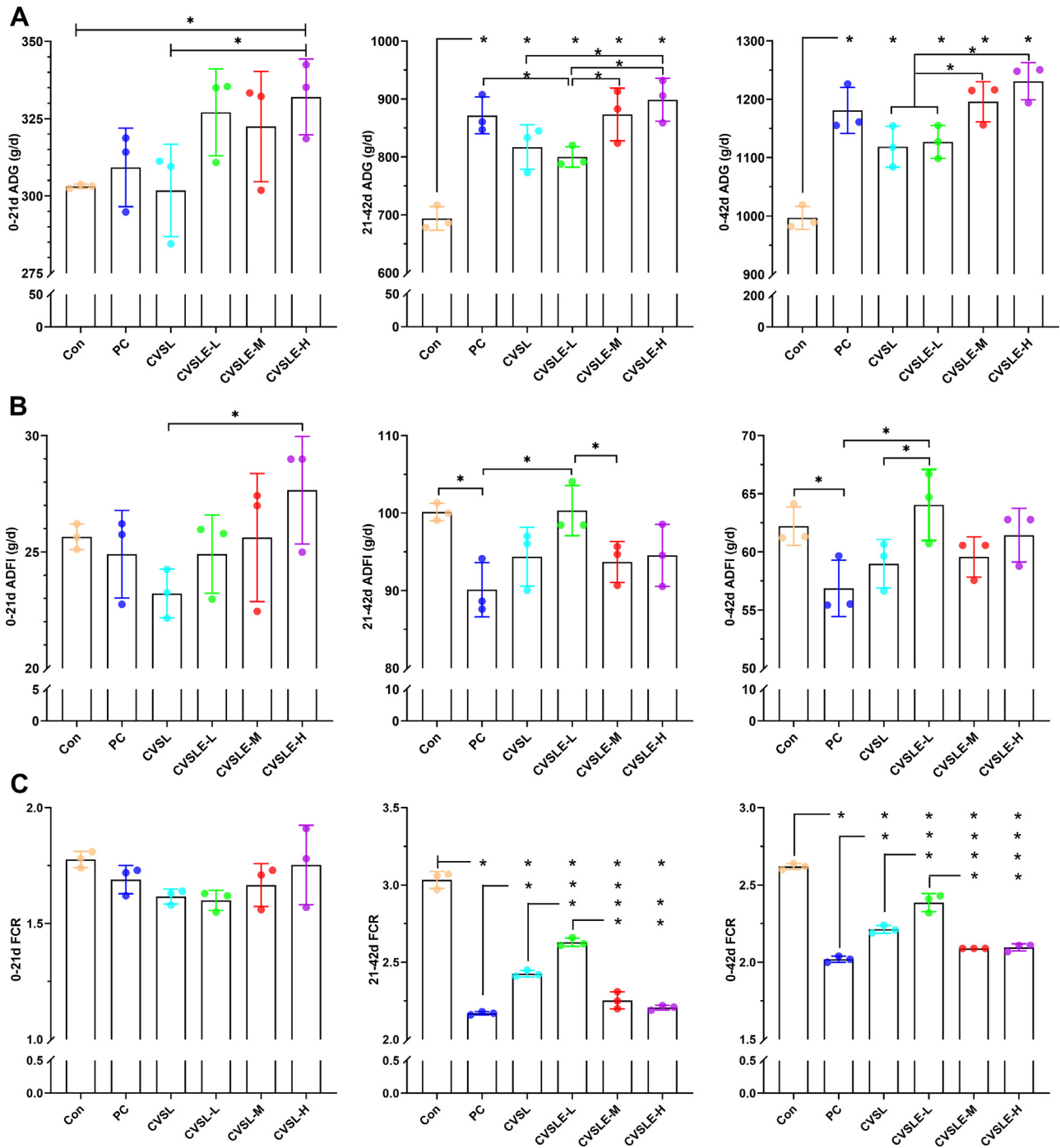


Figure 2. The effects of dietary CVSL and CVSLE feed on growth performance. The ADG (A), ADFI (B) and FCR (C) of different groups at 0-21d, 21-42d and 0-42d. Con, blank control; PC, Positive control; CVSL, 1% CVSL granules; CVSLE-L, 0.2% CVSLE granules; CVSLE-M, 0.5% CVSLE granules; CVSLE-H, 1.0% CVSLE granules. * $p < 0.05$.

illustrated in Fig. 4. At 21 d, the antibody titers of the positive control and CVSLE groups against NDV were significantly higher ($P < 0.05$) than those of the blank control group. The medium-dose CVSLE group exhibited the highest titer. Furthermore, the HI titers against AIV were significantly elevated in all treated groups ($P < 0.05$), with the positive control group demonstrating the highest titer. At 28 d, the antibody titers against NDV in the low and medium doses of the CVSLE groups were significantly enhanced ($P < 0.05$), while the HI titers against AIV were significantly increased in all treated groups ($P < 0.05$). At 42 d, the HI titers against

NDV in all CVSL and CVSLE groups were significantly higher than in the blank control group ($P < 0.05$). Only the medium dose of CVSLE demonstrated a higher titer against AIV.

Immune Organ Indices and Splenic Cytokines

The impact of dietary CVSL and CVSLE feed on immune function was reflected by organ indices and splenic cytokines (IL-2, IL-4, IL-10, and IFN- γ), as

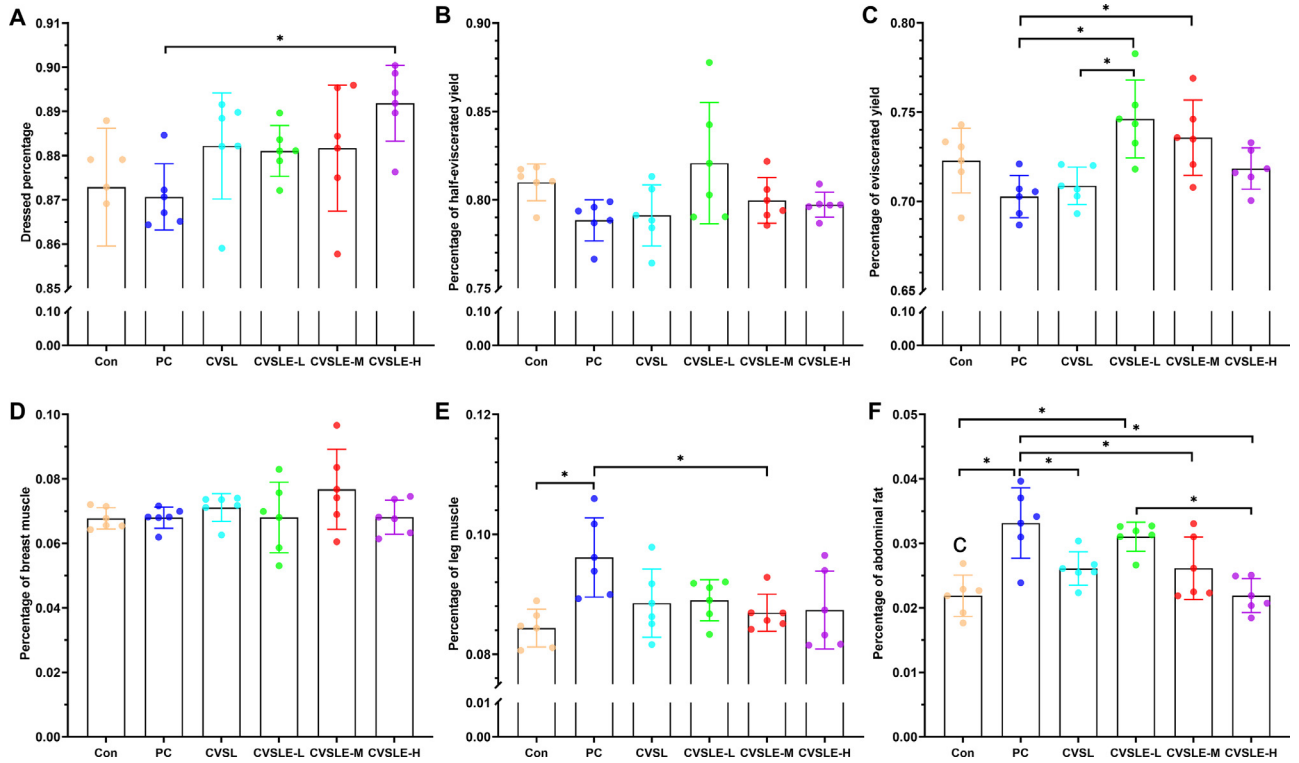


Figure 3. The effects of dietary CVSLE and CVSLE feed on slaughtering performance. (A) Dressed percentage; (B) Percentage of half-eviscerated yield; (C) Percentage of eviscerated yield; (D) Percentage of breast muscle; (E) Percentage of leg muscle; (F) Percentage of abdominal fat. Con, blank control; PC, Positive control; CVSLE, 1% CVSLE granules; CVSLE-L, 0.2% CVSLE granules; CVSLE-M, 0.5% CVSLE granules; CVSLE-H, 1.0% CVSLE granules. * $P < 0.05$.

illustrated in Figure 5. At 21 d, the spleen and bursa of Fabricius indices of the positive control and CVSLE groups were significantly increased ($P < 0.05$) in comparison to the blank control group, while no significant

differences were observed in the thymus index among all groups. At 28 d, the medium-dose CVSLE treatment resulted in a statistically significant enhancement of the spleen and bursa of Fabricius indices ($P <$

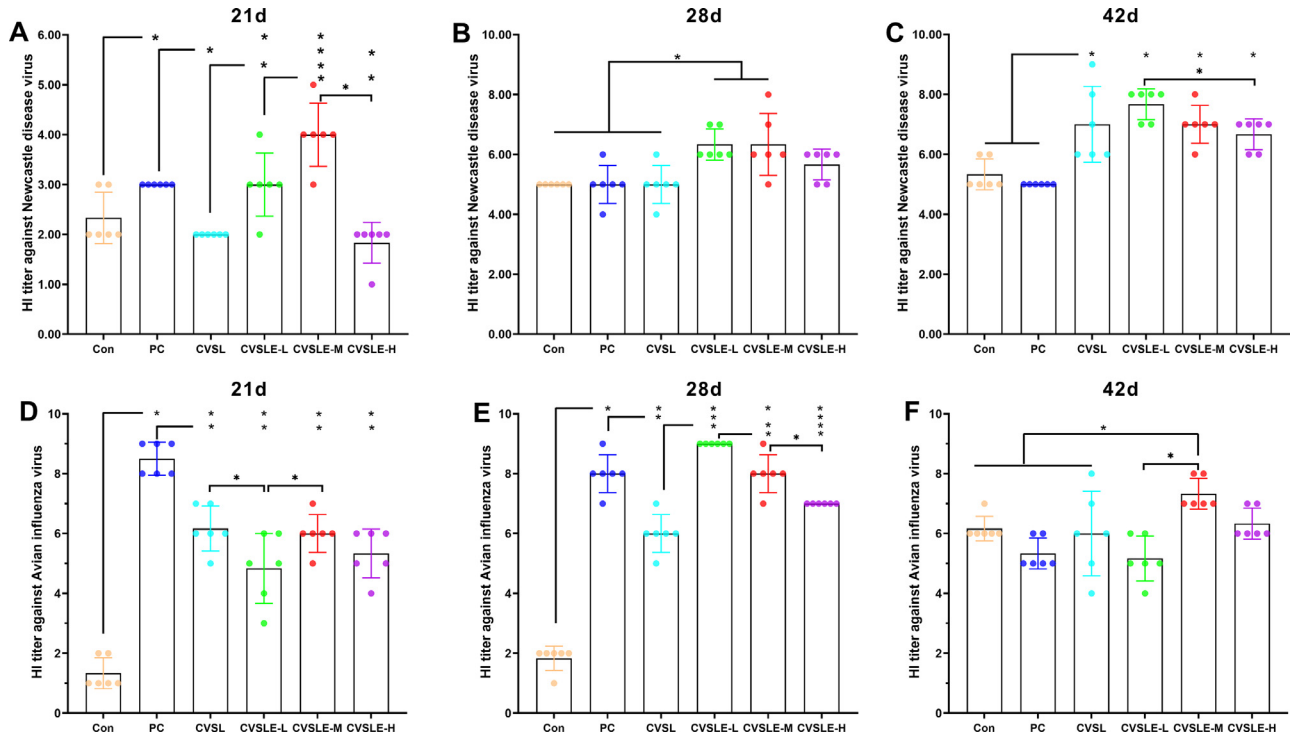


Figure 4. The HI antibody titers against NDV and AIV. The HI titers of different groups against NDV at 21d (A), 28d (B) and 42d (C). The HI titers of different groups against AIV at 21d (D), 28d (E) and 42d (F). Con, blank control; PC, Positive control; CVSLE, 1% CVSLE granules; CVSLE-L, 0.2% CVSLE granules; CVSLE-M, 0.5% CVSLE granules; CVSLE-H, 1.0% CVSLE granules. * $P < 0.05$.

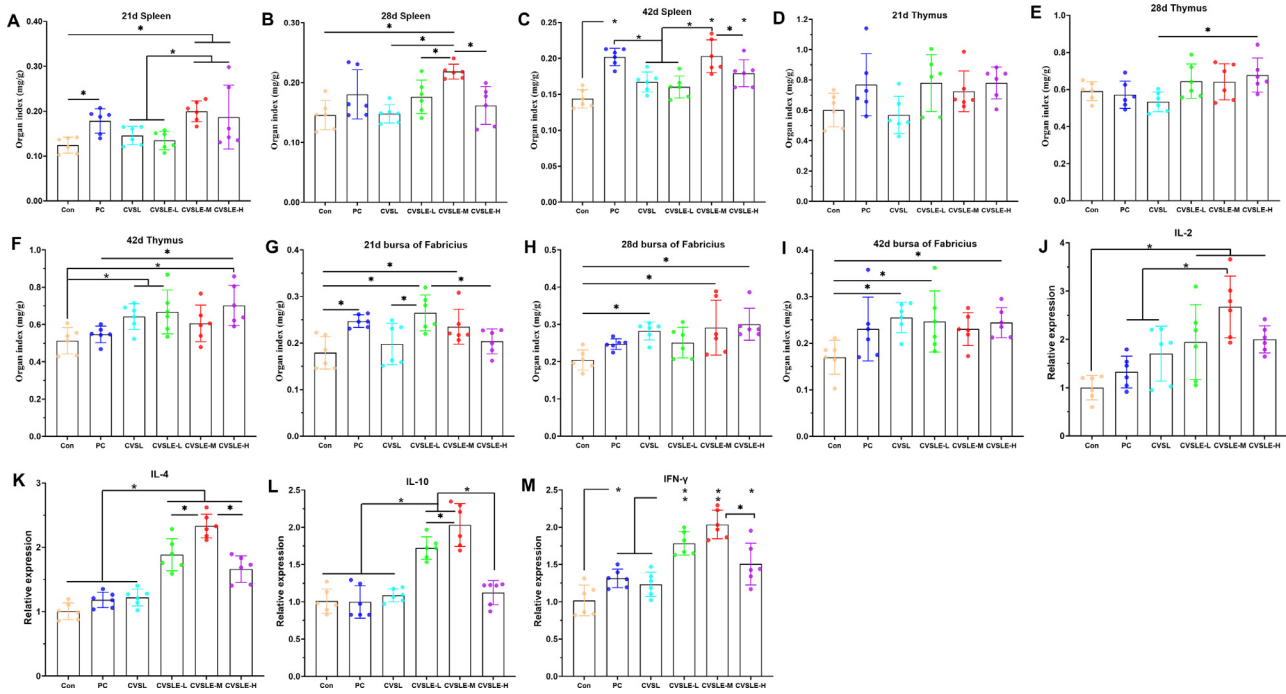


Figure 5. The effects of dietary CVSL and CVSLE feed on organ indices and splenic cytokines. (A) Spleen index at 21 d; (B) Spleen index at 28d; (C) Spleen index at 42 d; (D) Thymus index at 21d; (E) Thymus index at 28d; (F) Thymus index at 42d; (G) bursa of Fabricius index at 21d; (H) bursa of Fabricius index at 28d; (I) bursa of Fabricius index at 42d. The transcriptional levels of IL-2 (J), IL-4 (K), IL-10 (L) and IFN- γ (M) in spleen. Con, blank control; PC, Positive control; CVSL, 1% CVSL granules; CVSLE-L, 0.2% CVSLE granules; CVSLE-M, 0.5% CVSLE granules; CVSLE-H, 1.0% CVSLE granules. * $P < 0.05$.

0.05). At 42 d, the spleen index of the positive control group was significantly enhanced ($P < 0.05$), and CVSLE was observed to significantly elevate the spleen, thymus, and bursa of Fabricius indices ($P < 0.05$). The medium-dose CVSLE was observed to significantly elevate the levels of IL-2, IL-4, IL-10, and IFN- γ in comparison with the black control and positive control groups ($P < 0.05$).

Intestinal Barrier Function

The impact of dietary CVSL and CVSLE on intestinal barrier function is illustrated in Figure 6, including immune barrier (sIgA), physical barrier (ZO-1, OCL, and Muc-2), and flora barrier (*Salmonella*, *E. coli*, *Lactobacillus*, and *Bifidobacterium*). The levels of sIgA in the jejunum and ileum were significantly elevated in the positive control and medium-dose CVSLE groups when compared with the blank control group at d 21, 28, and 42 ($P < 0.05$). Similar outcomes were observed in the levels of ZO-1, OCL, and Muc-2, which were also significantly elevated by treatment with the positive control and CVSLE at d 21, 28, and 42 ($P < 0.05$). In the intestinal flora, treatment with the positive control, CVSL, and CVSLE resulted in a significant reduction in the amounts of *Salmonella* and *E. coli*. Conversely, the CVSL and CVSLE treatments led to a significant increase in the amounts of *Lactobacillus* and *Bifidobacterium* ($P < 0.05$). The medium-dose CVSLE demonstrated the most pronounced efficacy in enhancing intestinal barrier function.

DISCUSSION

Indeed, the use of antibiotics in modern broiler production is necessary to maintain growth performance and health due to their ability to prevent and treat infections and stressors that may be present in the environment. However, the need to reduce the use of antibiotics in order to maintain food security urgently requires more sustainable solutions for enhancement of broiler production. During the extraction process, both water and ethanol extraction methods were employed. Nevertheless, the ethanol extract was ultimately excluded from this study for the following reasons: (1) The yield of the water extraction was 12%, whereas the ethanol extraction yielded only 2% (data not shown), indicating a significantly lower yield in the ethanol extract. Additionally, the cost of water extraction was less than that of ethanol extraction. (2) Of particular importance is the widely accepted notion that the nutrients present in plants are primarily polysaccharides and proteins, which are water-soluble. It is widely acknowledged that the quantity of feed additives should not exceed 0.5% in typical circumstances. Accordingly, the 0.5% dose was established as the median, while the 0.2% and 1% doses were designated as the low and high doses, respectively. The findings of this study indicate that the 0.5% dose group exhibited the most pronounced growth-promoting effect on broilers.

Growth performance is the most important indicator and is directly related to the economic efficiency of broilers (Kithama et al., 2023). The present study showed growth performance was improved by CVSLE,

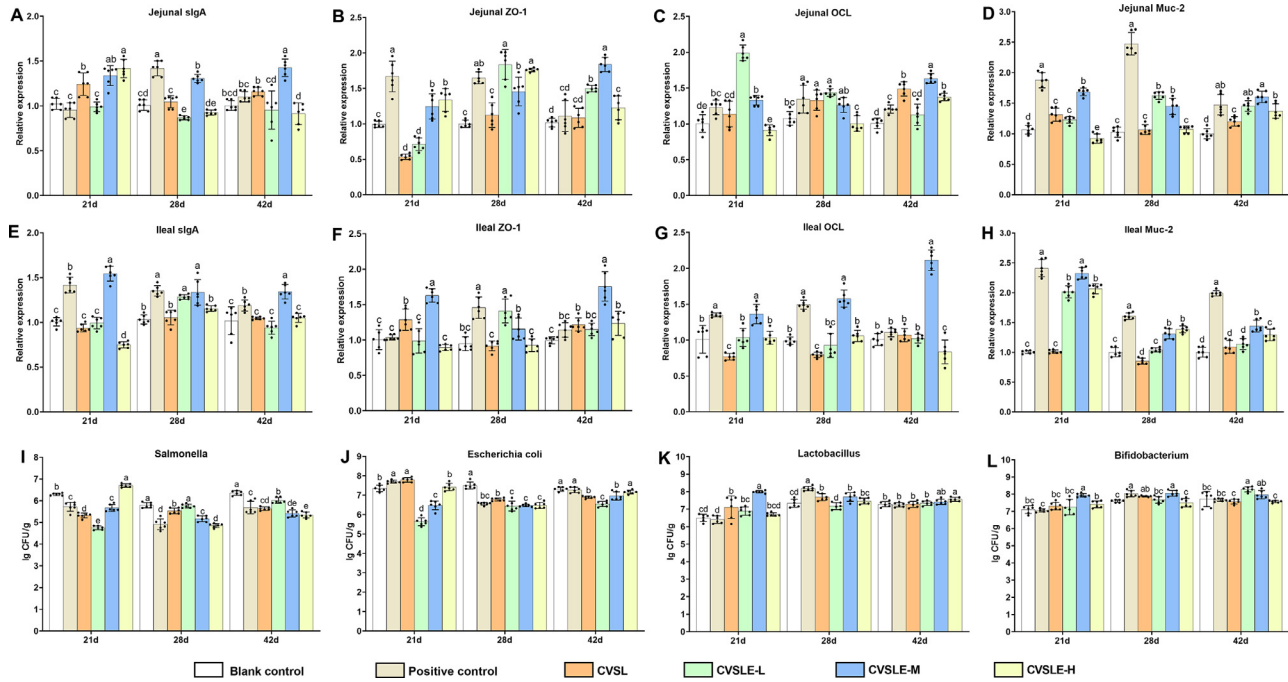


Figure 6. The effects of dietary CVSLE and CVSLE feed on intestinal barrier function. The transcriptional levels of sIgA (A), ZO-1 (B), OCL (C) and Muc-2 (D) in jejunum at 21, 28 and 42 d. The transcriptional levels of sIgA (E), ZO-1 (F), OCL (G) and Muc-2 (H) in ileum at 21, 28 and 42d. The amounts of *Salmonella* (I), *Escherichia coli* (J), *Lactobacillus* (K) and *Bifidobacterium* (L). Con, blank control; PC, Positive control; CVSLE, 1% CVSLE granules; CVSLE-L, 0.2% CVSLE granules; CVSLE-M, 0.5% CVSLE granules; CVSLE-H, 1.0% CVSLE granules. The different letter in column represents $P < 0.05$.

which was reflected by the increased ADG and ADFI and decreased FCR, which suggested the potential of the use of CVSLE as feed additives. Slaughter performance reflected carcass quality in poultry, which includes dressed percentage, half eviscerated yield, eviscerated yield, breast muscle percentage, abdominal fat percentage and so on. In general, a carcass yield of more than 80 percent and an evisceration yield of more than 60 percent indicate that broilers are producing good meat. In addition, breast muscle and leg muscle are the most valuable parts of a broiler carcass, and the higher the percentage of breast muscle and leg muscle, the higher the value of the broiler (Jespersen et al., 2021). The present study showed that there was no significant difference in the slaughter performance of broilers in comparison with the blank control group, suggesting that CVSLE and CVSLE had no significant effect on carcass quality.

It has been shown that herbs have a promoting effect on immune function by improving the indices of immune organs, humoral immunity and cellular immunity (Yunus et al., 2011). Immune organs are the basis of the body's immunity and play an important role in the performance of the body's immunity. Although immune cells are derived from bone marrow pluripotent stem cells, they need to differentiate and mature in the thymus, spleen and bursa of Fabricius before they can be transported to the peripheral immune organs or the whole body to perform cellular and humoral immune functions. Therefore, the developmental status of immune organs is related to the strength of the immune function in poultry and can be used to determine the immunity of the animal (Qui, 2022). The present study

showed significant improvement in immune organ indices of broilers in the CVSLE group, especially in the medium-dose group. Immunoglobulins refers to a class of globulin with antibody activity produced by the body in response to stimulation by antigenic substances and are capable of binding specifically to the antigen (Wang et al., 2021). The present study showed that the HI antibody titers of broilers in the CVSLE group were significantly elevated when compared to the blank control group, suggesting that CVSLE could enhance the resistant ability against AIV and NDV infections. Cytokines are mainly produced by antigen-activated lymphocytes and monocyte-macrophages, especially by helper T cells (Th). Th1 cells mainly secrete IL-2 and IFN- γ , and Th2 cells mainly secrete IL-4 and IL-10, which play a role in the immune network in terms of transmitting information, amplifying signals and enhancing immune effects (Viallard et al., 1999). The results of this study showed that the expressions of IL-2, IL-4, IL-10 and IFN- γ were significantly increased after CVSLE treatment, which was consistent with the increased antibody titers indicated that CVSLE could be an adjuvant to stimulate the production of cytokines, leading to increased levels of antibodies (Song et al., 2022). These results suggested that CVSLE could promote the development of immune organs and antibody titers by stimulating the production of cytokines, leading to enhance the immune function of broilers.

The intestine has a dual function as an important organ for digestion and absorption of nutrients, and as the first innate barrier to maintain the stability of the internal environment of the organism (Giromini et al., 2019). The intestinal barrier refers to the sum of

structures and functions that prevent harmful substances such as toxins and pathogenic microorganisms from entering intestinal mucosa, and includes mainly the physical, immune and flora barriers (Wan et al., 2022). The rich microflora is essential for the physiological function of the gut. Therefore, the composition and structure of the gut microbial community reflects the level of gut health. *Salmonella* is a pathogenic bacterium whose massive proliferation can lead to diarrhoea, chicken typhoid and chicken paratyphoid fever, etc., which can cause diarrhoea and death (Omar et al., 2021). *E. coli* are conditionally pathogenic bacteria, most of which are harmless and can help digestion. However, when it overpopulates or when the body's defences are weakened, it also causes high morbidity and mortality of chickens. *Lactobacillus* and *Bifidobacterium* can inhibit the reproduction of pathogenic bacteria in the intestinal tract to maintain the balance of the intestinal flora. The antagonism between pathogenic bacteria and probiotics forms the first line of defense of the intestinal tract, the biological barrier (Ji et al., 2022). The present study showed that the intestinal flora of broilers in the CVSLE group was significantly improved, with a significant reduction in *Salmonella* and *E. coli* and a significant increase in *Lactobacillus* and *Bifidobacterium*, especially in the medium-dose CVSLE group.

The second line of barrier, the immune barrier, is mainly secretory IgA (sIgA) which is secreted by lymphoid tissues in the lamina propria of the intestinal mucosa. sIgA can form an immunoprotected layer on the surface of the intestinal mucosa, preventing pathogens from attaching to the surface of intestinal cells through immune rejection (Jarosz et al., 2017). The present study showed that sIgA was significantly elevated in broilers in the both jejunum and ileum of medium-dose CVSLE group. The final barrier of the intestinal tract is the mechanical barrier, also known as the physical barrier, which is mainly composed of adherens and tight junctions, which are essential for maintaining the integrity of the intestinal tract and the polarity of the epithelial cells (Groschwitz and Hogan, 2009). It can prevent the pathogenic factors in the intestinal tract from entering the blood circulation. Among these, tight junctions are key molecules involved in the control of paracellular permeability of the intestinal barrier and maintaining the integrity of the epithelial structure. Tight junctions are multi-protein complexes that include a transmembrane barrier protein (occludin) and a peripheral scaffolding protein (ZO-1) (Proszkowiec-Weglarz et al., 2020). The function of occludin is to enter tight junctions, reduce the permeability of the membranes to resist small and large molecules (Shang et al., 2020). The present study showed that tight junction proteins were significantly elevated in the CVSLE group in the jejunum at 42 d and in the ileum at all test d. In summary, CVSLE can enhance the local immunity of the intestinal mucosa of broilers by increasing the content of sIgA, reduce the permeability of the intestinal mucosa by increasing the expression of tight junctions, and maintain the balance of the intestinal flora.

CONCLUSIONS

CVSLE exhibited positive regulatory effects on the growth performance, immune function, intestinal barrier, and carcass quality of broiler chickens. These findings suggest that CVSLE has the potential to be used as a feed additive for chickens, with the optimal dosage being 0.5% of the feed.

ACKNOWLEDGMENTS

This research was financially supported by the Natural Science Foundation of Sichuan Province (24NSFSC4908), Chengdu Science and Technology Project (2024-YF05-02140-SN) and the Program Sichuan Veterinary Medicine and Drug Innovation Group of China Agricultural Research System (SCCXTD-2020-18).

DISCLOSURES

The authors declare no conflicts of interest.

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